

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	10
Experiment Name	Data Collection and Initial Exploration

DESCRIPTION

To perform the experiment: Data Collection and Initial Exploration.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Glucose": [148, 85, 183, 89, 137],
    "BMI": [33.6, 26.6, 23.3, 28.1, 43.1],
    "Age": [50, 31, 32, 21, 33],
    "Outcome": [1, 0, 1, 0, 1]
})
print(df.head())
print("\nShape:", df.shape)
print("\nData types:")
print(df.dtypes)
print("\nSummary:")
print(df.describe())
print("\nOutcome counts:")
print(df["Outcome"].value_counts())
df[["Glucose", "BMI", "Age"]].hist(figsize=(8, 5))
plt.tight_layout(); plt.show()
```

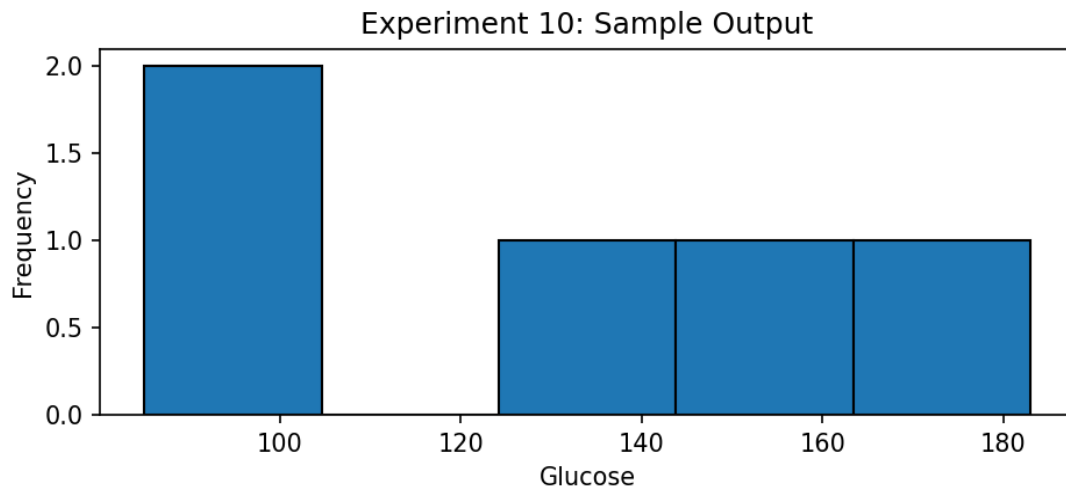
SAMPLE INPUT

Sample input: five example observations with glucose, BMI, age and outcome.

OUTPUT — SAMPLE CONSOLE OUTPUT

Shape: (5, 4)\nData types and descriptive statistics printed\nHistograms displayed.

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature